

Routing Configuration & Design

Overview

The enterprise network uses Open Shortest Path First (OSPF) as the primary dynamic routing protocol for communication between the Layer 3 core and distribution switches.

Static routing is also used where appropriate, including a default route toward the external/WAN network.

The routing design provides dynamic route learning and redundant paths between major network segments.

Routing Protocol

OSPF

OSPF is configured throughout the internal network to dynamically advertise and learn network prefixes.

OSPF allows the core and distribution devices to exchange routing information without requiring every route to be manually configured.

The routing tables show OSPF-learned routes using the O route designation.

For example, the B-side distribution switches learn the 10.6.0.0/24 network through OSPF with multiple available next hops.

OSPF Network Design

The network uses loopback interfaces on major Layer 3 devices.

Device	Loopback
R1	10.0.0.76/32
CSW1	10.0.0.77/32
CSW2	10.0.0.78/32
DSW-A1	10.0.0.79/32
DSW-A2	10.0.0.80/32

DSW-B1 10.0.0.81/32

DSW-B2 10.0.0.82/32

The loopbacks provide stable Layer 3 interfaces that can be advertised through OSPF.

The routing information confirms that these /32 loopback networks are present in the OSPF routing domain.

Routed Point-to-Point Links

The core and distribution infrastructure uses /30 IPv4 networks for point-to-point routed connections.

Examples include:

Network	Purpose
10.0.0.32/30	R1 ↔ CSW1
10.0.0.36/30	R1 ↔ CSW2
10.0.0.40/30	CSW1 ↔ CSW2
10.0.0.44/30	CSW1 ↔ DSW-A1
10.0.0.48/30	CSW1 ↔ DSW-A2
10.0.0.52/30	CSW1 ↔ DSW-B1
10.0.0.56/30	CSW1 ↔ DSW-B2
10.0.0.60/30	CSW2 ↔ DSW-A1
10.0.0.64/30	CSW2 ↔ DSW-A2
10.0.0.68/30	CSW2 ↔ DSW-B1
10.0.0.72/30	CSW2 ↔ DSW-B2

These links provide Layer 3 connectivity between the core and distribution devices.

OSPF Route Learning

OSPF allows the network devices to dynamically learn routes to remote VLAN networks.

For example, the B-side routing information shows:

10.3.0.0/24 → VLAN 10

10.4.0.0/24 → VLAN 20

10.5.0.0/24 → VLAN 30

as directly connected networks, while the 10.6.0.0/24 network is learned through OSPF.

This demonstrates the distinction between:

- Connected routes
 - OSPF-learned routes
 - Default routes
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Redundant Routing Paths

A major feature of the network design is routing redundancy.

The routing tables show multiple OSPF next hops for several destinations.

For example, the 10.6.0.0/24 network has two available paths:

10.0.0.57

10.0.0.73

Both paths have the same OSPF metric in the routing output.

This provides an alternate path if one of the routed links becomes unavailable.

The same routing design is reflected in the default route, which is also available through multiple next hops in the B-side routing information.

Default Route

R1 provides connectivity toward the external/WAN network.

The external network uses:

203.0.113.4/30

R1 uses the upstream address:

203.0.113.5

as the next hop for the default route.

The default route provides a path for traffic destined for networks outside the internal OSPF domain.

Routing Verification

The following Cisco IOS commands were used to verify routing functionality.

View the Routing Table

show ip route

This command displays connected, static, and dynamically learned routes.

Verify OSPF Neighbors

show ip ospf neighbor

This command verifies that OSPF neighbor relationships have been established between Layer 3 devices.

A successful neighbor relationship indicates that the devices are exchanging OSPF routing information.

Verify OSPF Configuration

show ip protocols

This command can be used to verify the active routing protocol, advertised networks, and routing parameters.

Verify OSPF Interfaces

show ip ospf interface

This command provides information about OSPF operation on individual interfaces.

Routing Troubleshooting

When troubleshooting routing problems, the following process can be used:

1. Verify the interface

show ip interface brief

Check that the interface has the correct IP address and is operational.

2. Verify the routing table

show ip route

Determine whether the destination network exists in the routing table.

3. Verify OSPF neighbors

show ip ospf neighbor

Confirm that expected OSPF adjacencies are established.

4. Verify OSPF configuration

show ip protocols

Confirm that the correct networks are being advertised.

5. Test connectivity

ping <destination>

Use ping to determine whether the destination is reachable.

6. Trace the path

tracert <destination>

Use tracert to identify where traffic stops along the path.

Routing Verification Checklist

Test	Command	Expected Result
Interface status	<code>show ip interface brief</code>	Required interfaces up/up
Routing table	<code>show ip route</code>	Expected routes present
OSPF neighbors	<code>show ip ospf neighbor</code>	Expected neighbors established
OSPF configuration	<code>show ip protocols</code>	OSPF enabled/configured
OSPF interfaces	<code>show ip ospf interface</code>	OSPF active on required interfaces
Connectivity	<code>ping</code>	Destination reachable
Path verification	<code>tracert</code>	Expected routing path

Routing Design Summary

The network combines OSPF, connected routes, and static/default routing to provide connectivity throughout the enterprise environment.

OSPF provides dynamic route exchange between the core and distribution layers, while /30 point-to-point networks provide efficient addressing for routed infrastructure links.

Loopback interfaces provide stable Layer 3 addresses for major network devices.

Multiple OSPF next hops provide routing redundancy and improve network resilience.

The overall routing design demonstrates practical experience with:

- OSPF
- Dynamic routing
- Static/default routing
- Layer 3 switching
- Point-to-point routing
- Loopback interfaces
- Route verification
- Routing troubleshooting
- Redundant network paths